

# Aidan Lee

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## EDUCATION

### Duke University

B.S. in Mathematics and Computer Science

Durham, NC

Expected May 2028

### Thomas Jefferson High School for Science and Technology

Advanced Studies Diploma — Weighted GPA: 4.3/4.0

Alexandria, VA

Graduated May 2025

## RESEARCH & WORK EXPERIENCE

### Software Engineering Research Assistant

QIAL Lab, Duke University

May 2026 – Jul 2026

Durham, NC

- Engineered an automated Python pipeline (ANTs/SimpleITK) that registered and propagated a **332**-region atlas across **27** brain MRIs, replacing manual per-region labeling
- Built a data-processing pipeline (pandas/NumPy) merging multimodal MRI, metadata, and behavioral tables into a unified analysis-ready dataset of **400+** features per mouse
- Automated an end-to-end data pipeline over **162** mice and **14** metrics, replacing manual analysis with parameterized, reproducible scripts that generate figures, tables, and reports

### Researcher / First Author

Emory University

Jun 2023 – Sep 2023

Atlanta, GA

- Improved stock-price forecasting accuracy by **30%** over a Facebook Prophet baseline (RMSE **27.4** vs. **39.3**) across **11** equities by implementing and tuning ARIMA models
- Published as first author at the 8th Intl. Conference on Research in Business, Management and Economics (Prague), advised by Dr. Karan Uppal

### Teaching Assistant

Math Enrichment Services

Jul 2024

McLean, VA

- Designed and taught a two-week competitive-math curriculum (AMC 8, MATHCOUNTS) to **30** students, delivering daily problem-solving lectures with quizzes and practice sets
- Administered end-of-week assessments and group review sessions targeting weak areas; top scorers improved by around **3** points over the two weeks

## PROJECTS

### MyWeb AI | FastAPI, PostgreSQL/pgvector, React/TypeScript, Chrome Extension (MV3), Claude API

Jul 2026

- Engineered a hybrid search engine fusing **BM25**, **384**-dim semantic vector similarity (pgvector + sentence-transformers), and trigram fuzzy matching into one weighted ranker over a **12**-table PostgreSQL schema
- Built a Manifest V3 Chrome extension and async FastAPI backend (**31** REST endpoints) ingesting **5+** content types — web pages, PDFs, YouTube transcripts, Google Docs/Sheets/Slides — with AI summaries, embeddings, and JWT + Google OAuth

### FencerHub | Next.js, TypeScript, Prisma/PostgreSQL, Cloudflare Stream

Jun 2026 - Jul 2026

- Architected a full-stack fencing training platform (Next.js, TypeScript, Prisma/PostgreSQL) serving **3** roles — fencers, coaches, parents — over a **20+**-table database with per-user access controls
- Developed a private-video system on Cloudflare Stream that delivers bout footage via signed access tokens, letting coaches pin timestamped feedback to specific moments

### DataCrawl | Gemini, LangGraph, FastAPI, Firebase

Mar 2026

- Engineered an AI-powered financial data pipeline converting natural-language requests into structured datasets from web sources via APIs, web scraping, and gated feeds
- Designed a **5+**-subagent LangGraph system (Gemini) for compliance, cost estimation, normalization, and schema validation; won **HackDuke 2026** award

### Fencing Visualizer | OpenCV, YOLOv11, React, FastAPI

Aug 2024 – May 2025

- Achieved **98.5%** mAP and **95%** multi-blade association accuracy by training a custom YOLOv11 model (transfer learning via Roboflow) on thousands of hand-annotated frames across **3** classes (blade, tip, guard)
- Built a markerless CV pipeline overlaying motion trails on uploaded footage — no tracking hardware needed — processing **30s** clips in under **3** min via a React/FastAPI app on AWS EC2

## ACTIVITIES & AWARDS

**Athletics:** NCAA Division I Varsity Fencing — Duke University [Foil, 2026 ACC Team Men's Fencing Champion]

**Activities:** Duke Asian Athletes Network (Chief Technology Officer)

**Awards:** USACO Gold Division — Python (Dec 2023), 3x AIME Qualifier (2022–2024)

## TECHNICAL SKILLS

**Languages:** Python, C++, Java, JavaScript, SQL, HTML/CSS

**Frameworks & Libraries:** React, Vite, FastAPI, Flask, LangGraph, PyTorch, OpenCV, Scikit-learn, pandas, NumPy, Matplotlib

**Tools & Cloud:** Git, Docker, AWS (EC2), Firebase, Arduino, VS Code